Using `markdown-zed` in the Terminal

Manual for MarkDown Live 0.7.1. MarkDown Live is one program, `markdown-zed`. Zed starts it in the background, but you can also use it yourself in the Terminal: to preview or print any Markdown file, to check your setup, or to repair the Zed keys. Each command below is followed by what you see.

Using markdown-zed in the Terminal

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1. Before you start

1.1 Where the program is

`./setup.sh` (in the MarkDown Live folder) installs it as `~/.cargo/bin/markdown-zed`. Check that the Terminal finds it:

```
markdown-zed --version
```

```
markdown-live 0.7.1
```

If you get `command not found: markdown-zed`, add the folder to your PATH once and try again:

```
echo 'export PATH="$HOME/.cargo/bin:$PATH"' >> ~/.zshrc
source ~/.zshrc
```

1.2 File names with spaces

Put quotes around paths with spaces (iCloud folders have them), or drag the file from Finder into the Terminal window, which adds the path for you:

```
markdown-zed preview "$HOME/Library/Mobile
Documents/com~apple~CloudDocs/Notes/plan.md"
```

1.3 All commands at a glance

```
markdown-zed --help
```

```
markdown-live 0.7.1

USAGE:
  markdown-zed [--stdio]                  language server (used by Zed)
  markdown-zed preview FILE [--line N] [--new-window]
  markdown-zed render FILE                print the rendered HTML
  markdown-zed serve                      standalone preview server
  markdown-zed doctor                     show the detected setup
  markdown-zed setup-zed [--key alt-m]    add the preview task + key binding to Zed
  markdown-zed print FILE                 open the preview and show the print dialog
  markdown-zed insert-slide FILE --line N insert <!-- slide --> (in Zed: through the
language server)

Environment: MARKDOWN_LIVE_SETTINGS (JSON, same keys as the Zed settings),
MARKDOWN_LIVE_CHROME
```

Command	Use it to
`preview`	open the live preview of a file
`print`	print a file, or save it as PDF
`insert-slide`	add a slide marker to a file
`render`	see the HTML MarkDown Live makes (for checking)
`doctor`	check what MarkDown Live found on your Mac
`setup-zed`	(re)create the Zed keys Option-M, Option-Shift-N, Option-Shift-P
`serve`	start a preview server by hand (normally automatic)

2. Preview a file

```
markdown-zed preview notes.md
```

```
MarkDown Live preview: http://127.0.0.1:58428/?t=4259a75a7deba16ef52fec95adcc5ea8
```

The preview opens in your browser and the address is printed. What it shows depends on whether Zed has the file open:

- **Zed has it open:** the preview shows what is in Zed, also text you haven't saved, and follows your cursor. This is exactly what **Option-M** does.
- **Zed doesn't:** a small preview server starts in the background and shows the file on disk. Every time the file is saved (by any editor) the preview updates. The server stops by itself two minutes after you close the preview.

Options:

Option	Meaning	Example
`--line N`	start at line N (scroll there)	`markdown-zed preview notes.md --line 120`
`--new-window`	always open a new browser window, even when a preview is already open	`markdown-zed preview notes.md --new-window`

Opening another file reuses the preview window that's already open: it switches to the new file.

3. Print a file or save it as PDF

```
markdown-zed print report.md
```

The preview opens (if it wasn't open yet) and shows the browser's print dialog for the document. Choose a printer, or **PDF → Save as PDF**. This is what **Option-Shift-P** does in Zed.

- The printout is in the light theme, without the preview's toolbar.
- A document with `<!-- slide -->` markers prints one slide per page: in **portrait** with the notes under the slide, in **landscape** with the notes on the next page.
- Videos are left out; charts and 3D molecules are printed as pictures.

If nothing seems to happen, the dialog is in the preview window behind other windows: bring the browser to the front.

4. Add a slide marker

```
markdown-zed insert-slide talk.md --line 12
```

Inserts a line `<!-- slide -->` so that it becomes line 12 of `talk.md` (the old line 12 moves down). If Zed has the file open, the change goes through Zed (so you can undo it with Cmd-Z); otherwise the file is changed on disk.

Before (lines 10–12):

```
- print it

## Questions?
```

After `--line 12`:

```
- print it

<!-- slide -->
## Questions?
```

5. Check your setup: `doctor`

```
markdown-zed doctor
```

Example on a Mac:

```
markdown-live 0.7.1
browser       : system default (no Chromium/Chrome found)
libraries     : bundled (50 embedded files)
instances     : /Users/duggie/Library/Caches/markdown-live/instances
running       : port 58428 (lsp), 1 open file(s)
math          : katex
plantuml      : local (/opt/homebrew/opt/openjdk/bin/java -jar
/opt/homebrew/opt/plantuml/libexec/plantuml.jar)
  java        : /opt/homebrew/opt/openjdk/bin/java
  jar         : /opt/homebrew/opt/plantuml/libexec/plantuml.jar
kroki         : https://kroki.io (only for ```kroki-…``` blocks)
ffmpeg        : /opt/homebrew/bin/ffmpeg
code chunks   : enabled (python: /Users/duggie/.venvs/mypythonenv/bin/python3)
```

How to read it:

Line	Meaning	If it's not what you expect
`browser`	where the preview opens	install Chromium/Chrome for a separate app window, or set `"browser"`
`running`	preview servers now running: **lsp** = started by Zed, **standalone** = started from the Terminal	a *standalone* one while working in Zed: `pkill -f "markdown-zed serve"`
`plantuml`, `java`, `jar`	how PlantUML diagrams are drawn	`brew install plantuml`, or set `"plantumlJar"` / `"java"`
`ffmpeg`	converts videos the browser can't play	`brew install ffmpeg`
`code chunks`	whether `{cmd=true}` code runs, and with which Python	set `"enableScriptExecution"` and `"python"` in Zed's settings

`doctor` uses the same settings as Zed (the `lsp` → `markdown-live` → `settings` part of Zed's settings.json).

6. Repair the Zed keys: `setup-zed`

```
markdown-zed setup-zed
```

```
updated /Users/duggie/.config/zed/tasks.json
updated /Users/duggie/.config/zed/keymap.json
Done. In Zed, open a Markdown file and press Alt-M to open the live preview.
  Alt-Shift-N → MarkDown Live: Insert new slide
  Alt-Shift-P → MarkDown Live: Print
Or: command palette → "task: spawn" → "MarkDown Live: Preview".
```

(Alt is the Option key.) It only changes its own part of those files, between the lines `// >>> markdown-live` and `// <<< markdown-live`; your other keys and tasks stay as they are. To use another key for the preview:

```
markdown-zed setup-zed --key alt-k
```

`./setup.sh` runs this for you after installing a new version.

7. See the HTML: `render`

```
markdown-zed render notes.md
```

For a file with

```
# Hello

Some *text*.
```

it prints:

```
<h1 id="hello" data-sourcepos="1:1-1:7">Hello<a href="#hello" aria-label="Link to
heading 'Hello'" data-heading-content="Hello" class="anchor"></a></h1>
<p data-sourcepos="3:1-3:12">Some <em data-sourcepos="3:6-3:11">text</em>.</p>
```

This is for checking what MarkDown Live makes of your Markdown (or for reporting a problem). It is not a finished web page: diagrams, charts and formulas are only drawn in the preview. To keep it in a file: `markdown-zed render notes.md > notes.html`. PlantUML diagrams and code chunks are run while rendering, so it can take a moment.

8. Settings from the Terminal

From the Terminal, `markdown-zed` uses your Zed settings (`lsp` → `markdown-live` → `settings` in Zed's settings.json). To change a setting for one command, put `MARKDOWN_LIVE_SETTINGS` in front of it, with the settings in JSON:

```
# dark preview, just this once
MARKDOWN_LIVE_SETTINGS='{"theme":"dark"}' markdown-zed preview notes.md

# run code chunks without asking, with another Python
MARKDOWN_LIVE_SETTINGS='{"enableScriptExecution":true,"python":"/usr/bin/python3"}'
markdown-zed render analysis.md > out.html
```

`MARKDOWN_LIVE_CHROME=/path/to/browser` picks the Chromium-type browser for the preview window. Note: a preview server that is already running keeps its settings; stop it first (`pkill -f "markdown-zed serve"`) if a setting doesn't seem to apply.

9. Preview servers

You normally never start one yourself, but it helps to know what runs:

Server	Started by	Settings	Shows
lsp	Zed, when you open a Markdown file	Zed's settings	your text in Zed, also unsaved
standalone	`markdown-zed preview/print` when Zed doesn't have the file open (or `markdown-zed serve`)	Zed's settings.json + `MARKDOWN_LIVE_SETTINGS`	the file on disk

The footer of the preview says which one you're looking at (**live (editor)** or **standalone**). Useful commands:

```
pgrep -fl markdown-zed              # what is running
pkill -f "markdown-zed serve"       # stop the standalone servers
```

To restart the one Zed uses: in Zed, command palette (Cmd-Shift-P) → **editor: restart language server**.

10. Recipes

Preview a file from Finder's folder in the Terminal

```
cd ~/Documents/Notes
markdown-zed preview plan.md
```

Make a PDF of a document

```
markdown-zed print report.md
# in the dialog: PDF → Save as PDF
```

Check a setup problem and send the details to someone

```
markdown-zed doctor > ~/Desktop/markdown-live-doctor.txt
```

Open the preview at the line you're working on, from a script

```
markdown-zed preview chapter3.md --line 250
```

Install a new version (from the MarkDown Live folder)

```
git fetch ../markdown-live-0.x.y.bundle 'refs/tags/*:refs/tags/*'
git checkout 0.x.y
./setup.sh
pkill -f "markdown-zed serve"
# then in Zed: Extensions → Rebuild MarkDown Live, and "editor: restart language
server"
```

11. Problems

Message or symptom	Fix
`command not found: markdown-zed`	see 1.1 (PATH)
`no such file: …`	check the path; put quotes around paths with spaces
`could not start the preview server`	run `markdown-zed doctor`; another program may block local ports
preview shows an old version of the file	it's a standalone server showing the file on disk: save the file, or open it in Zed
"code execution is off" or the wrong Python	`markdown-zed doctor` → `code chunks`; fix the settings, then `pkill -f "markdown-zed serve"`
the print dialog doesn't appear	it is in the browser window; bring it to the front

See also the manual *Writing Markdown in Zed with MarkDown Live* (`manual-markdown.md`).